

Circuits by Evidence

ChatGPT edition / Investigation 2

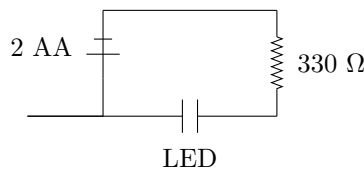
Three compact labs: path, resistance, and energy rate

THIS BOOK HAS ONE SOURCE CLASS

Use only isolated battery power, **12 V DC maximum**. For the procedures below, use the stated AA-cell holder. Never substitute a wall adapter or bench supply. Disconnect the holder before rewiring. If a cell, wire, or resistor becomes warm: disconnect, move hands away, tell the adult, and do not reuse it.

Lab A — What does “complete path” mean?

Build: two AA cells, a $330\ \Omega$ resistor and an LED in series. Try each change with power disconnected: reverse LED; remove one connection; bypass the resistor **only as a prediction—do not connect it**.



A source supplies potential difference; current needs a closed conducting path. The resistor limits current. The diagram shows connection, not tabletop shape.

Lab B — Make an I - V evidence table

Use **one** $220\ \Omega$, $0.25\ \text{W}$ resistor. Measure resistance with the battery disconnected. Then use 1, 2, 3 and 4 AA cells and measure voltage *across* the resistor. Measure current only after the adult verifies that the red lead is in the correct current jack and the meter is inserted in series. Connect for at most five seconds per reading.

Cells	V across (V)	I (A)	V/I (Ω)	$P = VI$ (W)
1				
2				
3				
4				

POWER RATING

Before the four-cell trial, calculate $P = V^2/R$ using the measured resistance and expected battery voltage. Proceed only if the result is below $0.25\ \text{W}$.

Lab C — Series and parallel are connection claims

Using two $1\ \text{k}\Omega$ resistors, predict and then measure equivalent resistance with **all batteries removed**. Series should exceed either resistor; parallel should be less than either. Explain the result as available paths, then check with $R_s = R_1 + R_2$ and $1/R_p = 1/R_1 + 1/R_2$.

Evidence record	
Prediction	_____
Changed variable	_____
Measured result	_____
Claim + because	_____

Read the graph, not just the formula

Plot I vertically and V horizontally. If the points lie near a straight line through zero, $V = IR$ is a useful model over this range. A slope changing with temperature would mean “constant R ” failed—not that Ohm’s law is a magic rule every component must obey.

Units tell a story

$1\text{ A} = 1\text{ C/s}$, $1\text{ V} = 1\text{ W/A}$, and $1\text{ }\Omega = 1\text{ V/A}$. Power $P = VI$ is an energy-transfer rate.

Evidence: NIST S04; DOE S06; DOE S07. Safety boundary: OSHA S01–S03. See the provider-neutral electricity source register.

Telos. Electricity & Magnetism — ChatGPT — Circuits by Evidence. Revised 2026-07-26. Project-created text and diagrams are licensed CC BY 4.0. Incorporated public-domain artwork remains public domain and is credited on the plate it appears with. See LICENSE and CREDITS.md in the source. Nothing here is professional advice; verify current regulations and follow the stated safety procedure.